THE CATHOLIC DIOCESE OF SAME
KINDOROKO GIRLS' SECONDARY SCHOOL
FORM FOUR LAST SECOND PRE–MOCK EXAMINATIONS

031/1 **PHYSICS 1**

(For Both Private and School Candidates)

Time: 2:30 Hours **Year: 2024**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section C carries **thirty (30)** marks.
4. Non – programmable calculators may be used.
5. Except for drawings which must be drawn in pencil, all writings must be in **blue** or **black** ink.
6. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
7. Write your **Examination Number** on every page of your answer booklet(s).
8. Where necessary the following constants may be used:
   Acceleration due to gravity, $g = 10m/s^2$
   Density of water, $\rho = 1\ gcm^{-3}$
   Pie, $\pi = 3.14$

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KINDOROKO PRE MOCK3 2024

SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the items (i) – (x) choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet(s) provided.

(i) A bus carrying a very big load on its carriers can easily overturn because:

A It cannot run fast
B Its equilibrium is stable
C Its centre of gravity is low
D It's more stable and can run fast
E Its centre of gravity is high

(ii) When charging body by friction, the particles which are transferred are:

A Protons
B Nuclei and protons
C Nuclei
D Electrons
E Protons and electrons

(iii) Which fact is most true about nature of Cathode rays:

A A stream of energy
B Used to examine bones
C A stream of molecule
D Stream of electrons
E Stream of gamma rays

(iv) The following will affect the rate of evaporation of water in a dam, EXCEPT:

A Surface area
B Depth
C Humidity
D Wind
E Temperature

(v) Electric pressure is measured in:

A Current
B Volts
C Watts
D Ampere
E Force

(vi) When a simple cell is in use the hydrogen bubbles are formed around a copper plate. This cell defect is called:

A Local action
B Polarization
C Amalgamation
D Hydrogenation
E Electrolysis

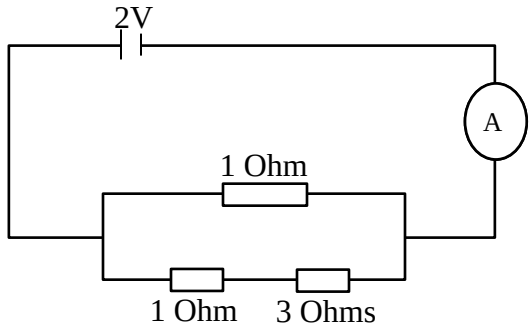
(vii) Which part of the camera functions as the retina in human eye?

A Shutter
B Diaphragm
C Film
D Convex lens
E Adjusting knob

(viii) A body moved upward a distance of 20m. Calculate the time taken to reach the maximum height.

A 5 seconds
B 10 seconds
C 15 seconds
D 2 seconds
E 20 seconds

(ix) In the circuit shown in Figure 1, the battery and the ammeter have negligible internal resistance. What will be the ammeter reading?



A 0.4 A
B 0.5 A
C 1.6 A
D 2.0 A
E 2.5 A

(x) The note from a plucked guitar will have a low pitch if the string is:

A Thick and long
B Thick and slack
C Thin and slack
D Thin and short
E Thick and short

2. Match the cases of the sound waves in List A with their corresponding parts of the sound waves terms in List B by writing the letter of the correct response beside the item number in your answer sheet provided.

LIST A	LIST B
(i) A range of the frequency of the sound that can be detected by a human ear	A Ultrasonic sound
	B Loudness
(ii) The sound with frequency below 20Hz	C Audibility range
(iii) The sound with frequency above 20KHz	D Ultra violet light
(iv) Applied in determining speed of sound	E Echo
(v) Effect occurs in designed halls and Auditoriums	F Reverberation
(vi) Applied in resetting an instrument to a given note	G Tuning fork
	H Infrasonic

SECTION B (54 Marks)

Answer **all** questions in this section

3. (a) Draw a labeled diagram of bulb and stem hydrometer used for measuring the relative density of liquid over the range 0.80 to 1.10. Mark graduation and show by an arrow the mark which it sinks when placed in water.

(b) An ordinary hydrometer of mass 27g floats with 4cm of its stem out of water. The area of cross section of stem is 0.75cm. What is total volume of the hydrometer?

(c) Briefly explain why hydrometer:

(i) Is weighed with lead shots

(ii) Has a narrow stem

4. (a) A mechanical engineer prefers to use long spanner in losing a tight nut than a short one.
   (i) State the principle that govern the process
   (ii) Explain why engineer choose to do so in 4(a) above
   (b) The valve of a cylinder containing 12kg of compressed gas is opened and the cylinder empties in 1 min 30 sec if the gas issues from the exit nozzle with an average velocity of 25m/s, find the force exerted on the cylinder.

5. (a) Briefly give reasons for the following:
   (i) The fundamental frequency may alter during the day
   (ii) Bat is able to fly at night without colliding with the wall
   (iii) Two distinct sounds are heard by the person standing on a railway line when sounded at distance.
   (b) The frequency obtained from a plucked string is 400Hz when the tension is 2N. Calculate:
   (i) The frequency when the tension is increased to 8N.
   (ii) The tension needed to produce a note frequency of 600Hz.

6. (a) An electric iron is labeled 240V, 1000W. What does this mean? Find the total cost of running simultaneously refrigerator rated 400W and a radio set rated 60W for 5 hours if electricity costs Tshs. 300 per KWh
   (b) A cell supplies current of 0.6A through a 2Ω coil and a current of 0.2A through a 7Ω Coil. Calculate the e.m.f and internal resistance of the cell.

7. (a) Explain the following observations:
   (i) Cork is placed between the double walls of the refrigerating chamber.
   (ii) A good cooking vessel should be black on the outside and not shiny white.
   (b) Differentiate between boiling and evaporation. Provide four (04) points.
   (c) If a radioactive isotope has a half-life of 2.5 hours, how long will it take for 256grams of the isotope to decay to 32 grams?

8. (a) (i) Explain why it would not be practical to make a Fortin type barometer with water.
   (ii) Explain the mode of action of hydraulic brakes.
   (b) (i) Elaborate the observation that when a mirror is cleaned with a dry cloth in a very dry atmosphere, the dust from the cloth stick to the glass.
   (ii) Three capacitors of values 2μF, 3μF, and 6μF are connected in parallel to a given circuit. What will be the capacitance of single chip capacitor to replace the above combination?

SECTION C (30 Marks)

Answer **two (2)** questions in this section

9. (a) An effort of 50N is applied to the handle of the screw jack of a car. The handle is 15cm long and there are 25 pitches in a centimeter. Determine the load rose if efficiency of the machine is 60%.
   (b) Distinguish between Mutual induction and self-induction as they are used in electromagnetism.
   (c) The hydrometer is immersed in liquid A and then in liquid B. The extent to which sinks in liquid B is less than liquid A. What conclusion can you draw from such observation?

10. (a) How long will a passenger sitting near the window of an SGR train travelling at 360km/h see a local train passing by in opposite direction with a speed of 100km/h? The length of the local train is 315m. Give the answer in seconds.

    (b) A Geger Muller tube records some count rates even though there is no radioactive source in the tube, state any three possible causes of those counts in a tube.

    (c) Boats use "SONAR" to detect the depth of water in the seas or lakes. A sonar pulse sent out by a boat arrives back after 3 seconds. If the speed of sound in water is 1500m/s, how deep is the lake/seas?

11. (a) An object 20mm high stands on the axis of a concave mirror of focal length of 100mm and at a distance of 300mm from the mirror. Draw a scale diagram to find the position, size and nature of the image produced.

    (b) Mr. Chakupewa observed that most of the car headlamps have parabolic metal reflector. Explain why do the car have that parabolic metal reflector

    (c) (i) Explain why primary and secondary coils of induction are wound on each other?

    (ii) Tanzania National Electric Supply Company (TANESCO) wishes to transfer electricity from a power plant to the city. Why high voltage is preferred to transfer electricity than low voltage?

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